

Eigenvalues of a coupled system of thermostat-type via a Birkhoff–Kellogg type Theorem

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Abstract

In this paper, by means of Birkhoff–Kellogg type Theorem in cones we address the existence of eigenvalues and the corresponding eigenvectors to a family of coupled system of thermostat type. The system is characterized by the presence of a real parameter that influences not only the differential equations but also the boundary conditions. Motivated by models of temperature regulation and feedback-controlled systems, we reformulate the original boundary value problems into systems of Hammerstein integral equations. The theoretical results are applied to three different classes of boundary conditions in $t = 0$, which are supported by examples.

1 Introduction

The study of coupled systems of second-order ordinary differential equations (ODEs)

$$-u_i''(t) = f_i(t, u_1(t), u_2(t)), \quad t \in (0, 1), \quad i = 1, 2,$$

has become a cornerstone in nonlinear analysis, mostly because it can be used to describe a wide variety of physical and biological processes. An extremely useful and widely-known method of establishing the existence of positive solutions is to rewrite the original system of boundary value problems (BVPs) as a system of perturbed Hammerstein integral equations, which is then studied by topological fixed-point techniques in cones (see [1, 5, 8, 17–20, 22, 23]). The depth of this area can be seen in the fact that an impressive diversity of boundary conditions has been pursued with success.

A progressive generalization of the boundary structures has been shown in the literature. Systems with multi-point and four-point coupled BCs are included in fundamental work. For example, Asif and Khan [3] investigate the problem that involved non-homogenous conditions of the form $x(1) = \alpha y(\xi)$ and $y(1) = \beta x(\eta)$. The Guo–Krasnosel'skiĭ cone-expansion and compression theorem was used to prove their existence. This classical theorem was also used by Henderson and Luca [14] in the study of a coupled system with multi-point boundary conditions.

The addition of nonlinear and functional boundary conditions marked a major advancement in the generalization process. A flexible framework based on the fixed-point index was developed by Infante and Pietramala [18, 20] to handle systems with fairly general nonlocal and nonlinear BCs, such as those of the type $u_i(0) = H_i[u_j]$ and $u_i(1) = G_i[u_j]$. Similar techniques were used by Goodrich [12] for systems with the nonlinear BCs $x(0) = H_1[\varphi_1]$ and $y(0) = H_2[\varphi_2]$. Similarly, Cui and Sun [9] examined singular superlinear systems with coupled integral BCs $x(1) = \alpha[y] = \int_0^1 y(t) dA(t)$, $y(1) = \beta[x] = \int_0^1 x(t) dB(t)$ and used cone-based techniques to demonstrate their existence.

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These models have gained additional levels of complexity with the passage of time. Infante and Pietramala [21] generalized their previous work to deal with impulsive BVPs where solutions displaying jumps guided by requirements like $\Delta u_i|_{t=\tau_i} = I_i(u_i(\tau_i))$. Another challenge arises when the nonlinearities involved derivatives. Xu and Zhang [36] did this by working in a C^1 space where they proved existence of an extended Stieltjes integral BCs coupled system by means of the spectral radius of a linear operator associated to it.

Although this rich literature covers a variety of boundary conditions, there are still gaps in the literature pertaining to systems where a parameter λ concurrently drives both differential equations and boundary conditions, this structure changes the problem into a nonlinear eigenvalue problem and called the spectrum problem; see Subsection 4.7 of [31] and the references therein. See also [2, 5, 19] for related work.

Cianciaruso, Infante and Pietramala [8, Section 3], by means of classical fixed point theory studied the following problem:

$$\begin{cases} u''(t) + g_1(t)f_1(t, u(t), v(t)) = 0, & \text{a.e. on } [0, 1], \\ v''(t) + g_2(t)f_2(t, u(t), v(t)) = 0, & \text{a.e. on } [0, 1], \\ u'(0) + H_{11}[u, v] = 0, \\ u(1) = \beta_1 u(\eta) + H_{12}[u, v], & 0 < \eta < 1, \\ v'(0) + H_{21}[u, v] = 0, \\ v(1) = \beta_2 v(\xi) + H_{22}[u, v], & 0 < \xi < 1, \end{cases} \quad (1)$$

In this paper, we address the existence of eigenvalues and the corresponding eigenvectors for a class of parameter-dependent coupled system of thermostat type similar to (1). Our motivation for studying such problems lies in their application in physical phenomena, such as modeling the problem of a cooling or heating system controlled by a thermostat. In these heat-flow problems, controllers, reacting to the sensors, are placed in specific points. These are widely studied problems in the context of linear [4, 6–8, 10, 11, 22, 23, 29, 32–35] and nonlinear [24–26, 30] controllers. In particular, we study the following system of ODEs

$$\begin{cases} -u_1''(t) = \lambda F_1(t, u_1(t), u_2(t)), & t \in (0, 1), \\ -u_2''(t) = \lambda F_2(t, u_2(t), u_1(t)), & t \in (0, 1), \end{cases} \quad (2)$$

subject to three families of coupled functional boundary conditions in which the parameter λ also appears:

1 - Dirichlet-type conditions in $t = 0$:

$$\begin{cases} u_1(0) = \lambda H_1[u_1, u_2], & \beta_1 u_1'(1) + u_1(\eta_1) = \lambda G_1[u_1, u_2], \\ u_2(0) = \lambda H_2[u_1, u_2], & \beta_2 u_2'(1) + u_2(\eta_2) = \lambda G_2[u_1, u_2], \end{cases} \quad (3)$$

2 - Neumann-type conditions in $t = 0$:

$$\begin{cases} u_1'(0) + \lambda H_1[u_1, u_2] = 0, & \beta_1 u_1'(1) + u_1(\eta_1) = \lambda G_1[u_1, u_2], \\ u_2'(0) + \lambda H_2[u_1, u_2] = 0, & \beta_2 u_2'(1) + u_2(\eta_2) = \lambda G_2[u_1, u_2], \end{cases} \quad (4)$$

3 - Mixed Neumann and Dirichlet type conditions in $t = 0$:

$$\begin{cases} u_1'(0) + \lambda H_1[u_1, u_2] = 0, & \beta_1 u_1'(1) + u_1(\eta_1) = \lambda G_1[u_1, u_2], \\ u_2(0) = \lambda H_2[u_1, u_2], & \beta_2 u_2'(1) + u_2(\eta_2) = \lambda G_2[u_1, u_2], \end{cases} \quad (5)$$

where $\beta_i > 0$, $\beta_i + \eta_i < 1$, $\eta_i \in (0, 1)$, H_i and G_i are suitable functionals. Here we use a Birkhoff-Kellogg type theorem in cones for the existence of eigenvalues for this family of parameter-dependent and coupled BVPs, a setting somewhat similar to the one employed by Infante in [16] in the context of elliptic systems. In particular, in Section 2 we prove a fairly general eigenvalue and eigenfunction existence result for a class of systems of Hammerstein integral equations (see Theorem 2.2), which covers, as special case the three types of the above mentioned families of systems of BVPs. In Section 3 we illustrate explicitly this fact, with the aid of three mathematical examples. Our results are new and complement the previous theory.

2 Eigenvalues for a system of Hammerstein integral equations

We first recall some useful ingredients. Let $(Z, \|\cdot\|)$ be a real Banach space, a cone $K \subset Z$ is a closed, convex set such that $\alpha K \subset K$ for all $\alpha \geq 0$ and $K \cap (-K) = \{0\}$. We consider the following sets

$$K_R = \{u \in K : \|u\| < R\}, \quad \overline{K}_R = \{u \in K : \|u\| \leq R\}, \quad \partial K_R = \{u \in K : \|u\| = R\},$$

where $R \in (0, +\infty)$. With these ingredients, we may recall the following Birkhoff-Kellogg type theorem on cones, due to Krasnosel'skiĭ and Ladyženskii.

Theorem 2.1 ([27, 28]). *Let $(Z, \|\cdot\|)$ be a real Banach space, let $\hat{S} : \overline{K}_R \rightarrow K$ be compact and suppose that*

$$\inf_{x \in \partial K_R} \|\hat{S}x\| > 0.$$

Then there exist $\lambda_0 \in (0, +\infty)$ and $x_0 \in \partial K_R$ such that $x_0 = \lambda_0 \hat{S}x_0$.

For the application of Birkhoff-Kellogg type theorem in cones, we make the following assumptions on the following system of Hammerstein integral equations. These assumptions are a special case of the ones in [8].

$$u_i(t) = \psi_{i0}H_i[u_1, u_2] + \psi_{i1}G_i[u_1, u_2] + \int_0^1 K_i(t, s)F_i(s, u_1(s), u_2(s))ds, \quad i \in \{1, 2\}, \quad (6)$$

(D₁) $F_i : [0, 1] \times [0, +\infty) \times [0, +\infty) \rightarrow [0, +\infty)$ hold the Caratheodory-type conditions

- (a) $F_i(\cdot, u_1, u_2)$ is measurable for each fixed u_1 and u_2 in $[0, +\infty)$,
- (b) $F_i(t, \cdot, \cdot)$ is continuous for a.e. $t \in [0, 1]$,
- (c) and for each $R > 0$, there exists $\psi_{iR} \in L^\infty[0, 1]$ such that

$$F_i(t, u_1, u_2) \leq \psi_{iR}(t) \quad \text{for all } u_1, u_2 \in (0, R) \text{ and a.e. } t \in [0, 1]$$

(D₂) The kernels $K_i : [0, 1] \times [0, 1] \rightarrow [0, +\infty)$ is measurable and for every $\bar{t} \in [0, 1]$, we have

$$\lim_{t \rightarrow \bar{t}} |K_i(t, s) - K_i(\bar{t}, s)| = 0$$

(D₃) For every $i = 1, 2$ there exist subintervals $[a_i, b_i] \subseteq [0, 1]$, functions $\Phi_i \in L^\infty[0, 1]$ and constants $c_i \in (0, 1]$ such that

$$K_i(t, s) \leq \Phi_i(s) \text{ for } t \in [0, 1] \text{ and a.e. } s \in [0, 1]$$

$$K_i(t, s) \geq c_i \Phi_i(s) \text{ for } t \in [a_i, b_i] \text{ and a.e. } s \in [0, 1]$$

(D₄) $\psi_{ij} \in C([0, 1], [0, +\infty))$ and there exists $c_{ij} \in (0, 1]$ such that

$$\psi_{ij}(t) \geq c_{ij}\|\psi_{ij}\| \text{ for every } t \in [a_i, b_i] \quad i = 1, 2, j = 0, 1.$$

We work in the product space $Y = C[0, 1] \times C[0, 1]$ with the norm $\|(u_1, u_2)\|_Y = \max(\|u_1\|_\infty, \|u_2\|_\infty)$, here $C[0, 1]$ is Banach space equipped with the infinity norm $\|u_i\|_\infty = \sup_{t \in [0, 1]} |u_i(t)|$. We consider a cone Q in Y , defined by

$$Q = \{(u_1, u_2) \mid u_i \in Q_i, \text{ for } i = 1, 2\},$$

where Q_i is the cone

$$Q_i = \{x \in C[0, 1] : x(t) \geq 0 \text{ for every } t \in [0, 1], \min_{t \in [a_i, b_i]} x(t) \geq \tilde{c}_i \|x\|_\infty\},$$

where $\tilde{c}_i = \min\{c_i, c_{i0}, c_{i1}\}$. We assume that

(D₅) $H_i, G_i : Q \rightarrow [0, +\infty)$ are compact functionals.

Under the assumptions (D₁)-(D₅), a routine check shows that the integral operator

$$S(u_1, u_2) := (S_1(u_1, u_2), S_2(u_1, u_2)),$$

where

$$\begin{pmatrix} S_1(u_1, u_2) \\ S_2(u_1, u_2) \end{pmatrix}(t) := \begin{pmatrix} \psi_{10}(t)H_1[u_1, u_2] + \psi_{11}(t)G_2[u_1, u_2] + \int_0^1 K_1(t, s)F_1(s, u_1(s), u_2(s))ds \\ \psi_{20}(t)H_2[u_1, u_2] + \psi_{21}(t)G_2[u_1, u_2] + \int_0^1 K_2(t, s)F_2(s, u_1(s), u_2(s))ds \end{pmatrix},$$

maps Q into Q and it is compact (see for example Lemma 1 in [17]). Note that

$$(u_1, u_2) \in \partial Q_R \implies \begin{cases} \|u_i\|_\infty \leq R \text{ for some } i \in \{1, 2\}, \\ \|u_i\|_\infty = R \text{ for every } i \in \{1, 2\}. \end{cases}$$

For the solvability of the system (2) with (3), (4), or (5) we now state and prove the following result.

Theorem 2.2. *Suppose that $R \in (0, +\infty)$ and the following conditions hold for $i = 1, 2$*

(1) *There exists $\gamma_{iR} \in C([0, 1], [0, +\infty))$ such that*

$$F_i(t, u_1, u_2) \geq \gamma_{iR}(t), \text{ for all } (t, u_1, u_2) \in [a_i, b_i] \times \prod_{k=1}^2 [\delta_{ik}\tilde{c}_i R, R],$$

where δ_{ik} is the classical Kronecker delta function.

(2) *There exist $\zeta_{iR}^H, \zeta_{iR}^G \in [0, +\infty)$ be such that*

$$H_i(u_1, u_2) \geq \zeta_{iR}^H \quad \text{and} \quad G_i(u_1, u_2) \geq \zeta_{iR}^G, \text{ for every } (u_1, u_2) \in \partial Q_R.$$

(3) *The inequality*

$$\sup_{t \in [a_i, b_i]} \left[\psi_{i0}(t)\zeta_{iR}^H + \psi_{i1}(t)\zeta_{iR}^G + \int_{a_i}^{b_i} K_i(t, s)\gamma_{iR}(s)ds \right] > 0 \quad (7)$$

holds.

Then there exist $\lambda_R \in (0, +\infty)$ and $(u_{1R}, u_{2R}) \in \partial Q_R$ such that $(u_{1R}, u_{2R}) = \lambda_R S(u_{1R}, u_{2R})$.

Proof. As we know, the operator S is compact. We need to prove,

$$\inf_{(u_1, u_2) \in \partial Q_R} \|S(u_1, u_2)\| > 0.$$

Take $(u_1, u_2) \in \partial Q_R$. Let us assume that

$$\|u_1\|_\infty = R \quad \text{and} \quad \|u_2\|_\infty \leq R.$$

Then we have

$$\|S(u_1, u_2)\| = \max\{\|S_1(u_1, u_2)\|_\infty, \|S_2(u_1, u_2)\|_\infty\} \geq \|S_1(u_1, u_2)\|_\infty. \quad (8)$$

Now we have

$$\begin{aligned} \|S_1(u_1, u_2)\|_\infty &= \sup_{t \in [0, 1]} \left\{ \psi_{10}(t)H_1[u_1, u_2] + \psi_{11}(t)G_1[u_1, u_2] + \int_0^1 K_1(t, s)F_1(s, u_1(s), u_2(s))ds \right\} \\ &\geq \sup_{t \in [a_1, b_1]} \left\{ \psi_{10}(t)H_1[u_1, u_2] + \psi_{11}(t)G_1[u_1, u_2] + \int_{a_1}^{b_1} K_1(t, s)F_1(s, u_1(s), u_2(s))ds \right\}. \end{aligned}$$

Now since $u_1 \in Q_1$ and $\|u_1\|_\infty = R$, we have

$$\tilde{c}_1 R \leq u_1(t) \leq R \quad \text{or} \quad 0 \leq u_2(t) \leq R, \quad \text{for every } t \in [a_1, b_1].$$

Thus we can use hypothesis (1) and (2).

$$\|S_1(u_1, u_2)\|_\infty \geq \sup_{t \in [a_1, b_1]} \left\{ \psi_{10}(t)\zeta_{1R}^H + \psi_{11}(t)\zeta_{1R}^G + \int_{a_1}^{b_1} K_1(t, s)\gamma_{1R}(s)ds \right\}. \quad (9)$$

On the other hand, if we assume that $\|u_1\|_\infty \geq R$ and $\|u_2\|_\infty = R$, reasoning as above, we obtain the inequality

$$\|S_2(u_1, u_2)\|_\infty \geq \sup_{t \in [a_2, b_2]} \left\{ \psi_{20}(t)\zeta_{2R}^H + \psi_{21}(t)\zeta_{2R}^G + \int_{a_2}^{b_2} K_2(t, s)\gamma_{2R}(s)ds \right\}. \quad (10)$$

In both cases, for every $(u_1, u_2) \in \partial Q_R$ we have, from (8), that

$$\|S(u_1, u_2)\| \geq \min_{i=1,2} \sup_{t \in [a_i, b_i]} \left\{ \psi_{i0}(t)\zeta_{iR}^H + \psi_{i1}(t)\zeta_{iR}^G + \int_{a_i}^{b_i} K_i(t, s)\gamma_{iR}(s)ds \right\}.$$

Note that the right-hand side of the above inequality is independent of (u_1, u_2) , which implies

$$\inf_{(u_1, u_2) \in \partial Q_R} \|S(u_1, u_2)\| > 0,$$

and the result follows by Theorem 2.1. □

3 Applications to coupled system of BVPs

We now proceed to apply Theorem 2.2 to the BVPs mentioned in the Introduction.

3.1 The BVP (2)-(3)

We begin with the BVP

$$\begin{cases} -u_1''(t) = \lambda F_1(t, u_1(t), u_2(t)), & t \in (0, 1), \\ -u_2''(t) = \lambda F_2(t, u_2(t), u_1(t)), & t \in (0, 1), \\ u_1(0) = \lambda H_1[u_1, u_2], & \beta_1 u_1'(1) + u_1(\eta_1) = \lambda G_1[u_1, u_2], \\ u_2(0) = \lambda H_2[u_1, u_2], & \beta_2 u_2'(1) + u_2(\eta_2) = \lambda G_2[u_1, u_2]. \end{cases} \quad (11)$$

The system (11) can be written in the integral form,

$$u_i(t) = \lambda \left[\psi_{i,0}(t) H_i[u_1, u_2] + \psi_{i,1}(t) G_i[u_1, u_2] + \int_0^1 K_i(t, s) F_i(s, u_1(s), u_2(s)) ds \right], \quad i \in \{1, 2\}. \quad (12)$$

Hereafter, for simplicity and with slight abuse of notation, we use the same notation for K_i , Φ_i , $\psi_{i,0}$ and $\psi_{i,1}$ for all BVPs. As in [4, 6, 23], the Green's functions associated with the system are given by:

$$K_i(t, s) = \frac{t}{\beta_i + \eta_i} \beta_i + \frac{t}{\beta_i + \eta_i} ((\eta_i - s) \mathbf{1}_{[0, \eta]}(s)) - (t - s) \mathbf{1}_{[0, t]}(s), \quad (13)$$

where $\mathbf{1}_{[a, b]}(x)$ is the indicator function, equal to 1 if $x \in [a, b]$ and 0 otherwise. With the choice of the subinterval $[a_i, b_i] \subset (0, \beta_i + \eta_i) \subset (0, 1)$ for $i \in \{1, 2\}$, the hypotheses (D_2) and (D_3) are satisfied. With $\Phi_i(s)$ is given by

$$\Phi_i(s) = \begin{cases} s, & \text{if } \beta_i + \eta_i \geq \frac{1}{2}, \\ \left[\frac{1 - (\beta_i + \eta_i)}{\beta_i + \eta_i} \right] s, & \text{if } \beta_i + \eta_i < \frac{1}{2}, \end{cases}$$

and the constant c_i is

$$c_i = \begin{cases} \min \left\{ \frac{a_i \beta_i}{\beta_i + \eta_i}, \frac{\beta_i + \eta_i - b_i}{\beta_i + \eta_i} \right\}, & \text{if } \beta_i + \eta_i \geq \frac{1}{2}, \\ \min \left\{ \frac{a_i \beta_i}{1 - (\beta_i + \eta_i)}, \frac{\beta_i + \eta_i - b_i}{1 - (\beta_i + \eta_i)} \right\}, & \text{if } \beta_i + \eta_i < \frac{1}{2}. \end{cases}$$

The functions $\psi_{i,0}(t)$, derived in [23], are given by:

$$\psi_{i,0}(t) = 1 - \frac{t}{\beta_i + \eta_i}, \quad (14)$$

Since $\psi_{i,0}$ are decreasing functions on $[0, 1]$, we have $\|\psi_{i,0}\|_\infty = \psi_{i,0}(0) = 1$. Moreover, for $t \in [a_i, b_i]$, we have:

$$\psi_{i,0}(t) \geq \psi_{i,0}(b_i) = 1 - \frac{b_i}{\beta_i + \eta_i}.$$

Thus, condition D_4 is satisfied with $c_{i0} = 1 - \frac{b_i}{\beta_i + \eta_i}$. since $b_i < \beta_i + \eta_i$, $c_{i0} \in (0, 1)$.

Both $K_i(t, s)$ and $\psi_{i,0}$ change sign when $\beta_i + \eta_i < 1$, but are positive on the strip $0 \leq b_i \leq t$, $b_i < \beta_i + \eta_i$. For a detailed analysis, we refer the reader to [4, 23].

The functions $\psi_{i,1}(t)$, calculated using the same methodology as [23], are:

$$\psi_{i,1}(t) = \frac{t}{\beta_i + \eta_i}, \quad (15)$$

$\psi_{i,1}(t) = \frac{t}{\beta_i + \eta_i}$ are non-negative and increasing on $[0, 1]$. We have $\|\psi_{i,1}\|_\infty = \psi_{i,1}(1) = \frac{1}{\beta_i + \eta_i}$. For $t \in [a_i, b_i]$, $\psi_{i,1}(t) \geq \psi_{i,1}(a_i) = \frac{a_i}{\beta_i + \eta_i}$. Thus, condition (D_4) is satisfied with $c_{i1} = \frac{\psi_{i,1}(a_i)}{\|\psi_{i,1}\|_\infty} = a_i$. Since $0 < a_i < 1$, $c_{i1} \in (0, 1)$.

Definition 3.1. We say that λ is an eigenvalue of the system (11), with a corresponding eigenfunction $(u_1, u_2) \in Q$ such that $\|(u_1, u_2)\| > 0$, if the pair $(\lambda, (u_1, u_2))$ satisfies the system of Hammerstein integral equations (12).

Now we can state the following existence result.

Theorem 3.2. Let $F_i : [0, 1] \times [0, +\infty) \times [0, +\infty) \rightarrow (0, +\infty)$ be continuous and $[a_i, b_i] \subset (0, \beta_i + \eta_i) \subset (0, 1)$. Let $\tilde{c}_i = c_i$ and $R \in (0, +\infty)$, further assume that the conditions (1)-(3) of Theorem 2.2 hold. Then there exist λ_R and $(u_{1R}, u_{2R}) \in \partial Q$ that satisfy the system (11).

We illustrate the applicability of the previous theorem in a specific example.

Example 3.3. Consider the system

$$\begin{cases} -u_1''(t) = \lambda \frac{1}{2} (u_1(t) + u_2^3(t) + 2), \\ -u_2''(t) = \lambda \frac{1}{2} (u_1^2(t) + u_2^2(t) + 1), \\ u_1(0) = \lambda \left(\frac{1}{12} u_1 \left(\frac{1}{3} \right) + \frac{1}{12} u_2(1) + \frac{1}{3} \right), \\ u_2(0) = \lambda \left(\frac{1}{6} u_1 \left(\frac{1}{3} \right) + \frac{1}{10} u_2(1) + \frac{1}{5} \right), \\ \frac{1}{4} u_1'(1) + u_1 \left(\frac{1}{4} \right) = \lambda \left(\frac{1}{2} \left(u_1 \left(\frac{1}{6} \right) \right)^{1/2} + \frac{\sqrt{2}}{20} \left(u_2 \left(\frac{1}{5} \right) \right)^3 \right), \\ \frac{1}{3} u_2'(1) + u_2 \left(\frac{1}{4} \right) = \lambda \left(u_1 \left(\frac{1}{3} \right) + u_2 \left(\frac{1}{3} \right) \right). \end{cases} \quad (16)$$

For $R \in (0, +\infty)$, we may take $[a_i, b_i] = \left[\frac{1}{6}, \frac{1}{3} \right]$, $\gamma_{1R}(t) = \frac{1}{2}(\tilde{c}_1 R + 2)$, $\gamma_{2R}(t) = \frac{1}{2}((\tilde{c}_2 R)^2 + 1)$, $\tilde{c}_1 = \frac{1}{12}$, $\tilde{c}_2 = \frac{1}{9}$, $\zeta_{1R}^H = 1/3 > 0$, $\zeta_{2R}^H = 1/5 > 0$, $\zeta_{1R}^G = 0$ and $\zeta_{2R}^G = 0$. Therefore, Condition 3 of Theorem 2.2 is satisfied for $i = 1$,

$$\begin{aligned} \sup_{t \in [\frac{1}{6}, \frac{1}{3}]} \left[\left(1 - \frac{t}{\beta_1 + \eta_1}\right) \frac{1}{3} + \frac{1}{2}(\tilde{c}_1 R + 2) \int_{\frac{1}{6}}^{\frac{1}{3}} K_1(t, s) ds \right] &\geq \\ \sup_{t \in [\frac{1}{6}, \frac{1}{3}]} \left[\left(\frac{5}{18}\right) + \frac{1}{24}(R + 2) \int_{\frac{1}{6}}^{\frac{1}{3}} c_1 s ds \right] &= \\ \sup_{t \in [\frac{1}{6}, \frac{1}{3}]} \left[\frac{5}{18} + \frac{1}{3456}(R + 2) \right] &> 0 \end{aligned}$$

and $i = 2$

$$\begin{aligned} \sup_{t \in [\frac{1}{6}, \frac{1}{3}]} \left[\left(1 - \frac{t}{\beta_2 + \eta_2}\right) \frac{1}{5} + \frac{1}{2}((\tilde{c}_2 R)^2 + 1) \int_{\frac{1}{6}}^{\frac{1}{3}} K_2(t, s) ds \right] &\geq \\ \sup_{t \in [\frac{1}{6}, \frac{1}{3}]} \left[\left(\frac{4}{35}\right) + \frac{1}{162}(R^2 + 1) \int_{\frac{1}{6}}^{\frac{1}{3}} c_2 s ds \right] &= \\ \sup_{t \in [\frac{1}{6}, \frac{1}{3}]} \left[\frac{4}{35} + \frac{3}{17496}(R^2 + 1) \right] &> 0, \end{aligned}$$

which implies that (7) is satisfied for every $R \in (0, +\infty)$.

Thus, we can apply Theorem 2.2 to obtain our desired solution.

3.2 The BVP (2)-(4)

The system of BVPs (2)-(4) can also be written in integral form (12).

As in [22, 23], the Green's functions associated with the system are given by:

$$K_i(t, s) = \beta_i + (\eta_i - s)\mathbf{1}_{[0, \eta_i]}(s) - (t - s)\mathbf{1}_{[0, t]}(s). \quad (17)$$

With the choice of the subinterval $[a_i, b_i] \subset (0, \beta_i + \eta_i) \subset (0, 1)$ for $i \in \{1, 2\}$, the hypotheses (D_2) and (D_3) are satisfied. With $\Phi_i(s)$ is given by

$$\Phi_i(s) = \begin{cases} \beta_i + \eta_i, & \text{if } \beta_i + \eta_i \geq 1/2 \\ 1 - (\beta_i + \eta_i), & \text{if } \beta_i + \eta_i < 1/2 \end{cases}$$

and the constant c_i is

$$c_i = \begin{cases} \frac{\beta_i + \eta_i - b_i}{\beta_i + \eta_i}, & \text{for } \beta_i + \eta_i \geq \frac{1}{2}, \\ \frac{\beta_i + \eta_i - b_i}{1 - (\beta_i + \eta_i)}, & \text{for } \beta_i + \eta_i < \frac{1}{2}. \end{cases}$$

From [23], we know that

$$\psi_{i,0}(t) = \beta_i + \eta_i - t \quad (18)$$

$$\text{with } \|\psi_{i,0}\| = \begin{cases} \beta_i + \eta_i & \text{for } \beta_i + \eta_i \geq \frac{1}{2} \\ 1 - (\beta_i + \eta_i) & \text{for } \beta_i + \eta_i < \frac{1}{2} \end{cases}.$$

$\psi_{i,0}$ are decreasing functions of t for $i \in \{1, 2\}$ and $\min_{t \in [a_i, b_i]} \psi_{i,0}(t) = \psi_{i,0}(b_i) = \beta_i + \eta_i - b_i$. This minimum is positive if $b_i < \beta_i + \eta_i$.

Therefore, the constant c_{i0} is:

$$c_{i0} = \frac{\min_{t \in [a_i, b_i]} \psi_{i,0}(t)}{\|\psi_{i,0}\|_\infty} = \frac{\beta_i + \eta_i - b_i}{\beta_i + \eta_i}.$$

In this case we calculated the functions $\psi_{i,1}$, as

$$\psi_{i,1} := 1,$$

which are non-negative on $[0, 1]$. We have $\|\psi_{i,1}\|_\infty = 1$. On the subinterval $[a_i, b_i]$, $\min_{t \in [a_i, b_i]} \psi_{i,1}(t) = 1$. Thus, the constant c_{i1} is:

$$c_{i1} = \frac{\min_{t \in [a_i, b_i]} \psi_{i,1}(t)}{\|\psi_{i,1}\|_\infty} = \frac{1}{1} = 1.$$

Now for the solvability of the system (2)-(4), we can state a result similar to Theorem 3.2.

Example 3.4. Consider the system

$$\begin{cases} -u_1''(t) = \lambda ((u_1(t))^2 + \sin^2(u_2(t)) + 1), \\ -u_2''(t) = \lambda (e^{u_1(t)} + u_2^3(t) + 1), \\ u_1'(0) + \lambda \left(\frac{1}{10}u_1(1) + \frac{1}{10}u_2(1) + \frac{1}{5} \right) = 0, \\ u_2'(0) + \lambda \left(\frac{1}{10}u_1\left(\frac{1}{2}\right) + \frac{1}{20}u_2(1) + \frac{1}{10} \right) = 0, \\ \frac{1}{4}u_1'(1) + u_1\left(\frac{1}{4}\right) = \lambda \left(\frac{1}{4} \left(\sqrt{u_1\left(\frac{1}{4}\right)} \right) + \frac{1}{8}(u_2(1)^2) + \frac{1}{5} \right), \\ \frac{1}{3}u_1'(1) + u_1\left(\frac{1}{4}\right) = \lambda \left(\frac{1}{6}u_1\left(\frac{1}{3}\right) + \frac{1}{6}u_2\left(\frac{1}{4}\right) + \frac{1}{6} \right). \end{cases} \quad (19)$$

For $R \in (0, +\infty)$, we may take $[a_i, b_i] = \left[\frac{1}{6}, \frac{1}{3}\right]$, $\gamma_{1R}(t) = (\tilde{c}_1 R)^2 + 1$, $\gamma_{2R}(t) = (\tilde{c}_2 R)^2 + 2$, $\tilde{c}_1 = \frac{1}{3}$, $\tilde{c}_2 = \frac{3}{7}$, $\zeta_{1R}^H = \frac{1}{5}$, $\zeta_{2R}^H = \frac{1}{10}$, $\zeta_{1R}^G = \frac{1}{5}$ and $\zeta_{2R}^G = \frac{1}{6}$. Therefore, Condition 3 of Theorem 2.2 is satisfied for $i = 1$,

$$\begin{aligned} \sup_{t \in [\frac{1}{6}, \frac{1}{3}]} \left[(\beta_1 + \eta_1 - t) \zeta_{1R}^H + \zeta_{1R}^G + \int_{\frac{1}{6}}^{\frac{1}{3}} K_1(t, s) \gamma_{1R}(s) ds \right] &\geq \\ \sup_{t \in [\frac{1}{6}, \frac{1}{3}]} \left[(\beta_1 + \eta_1 - b_1) \frac{1}{5} + \frac{1}{5} + \frac{(\tilde{c}_1 R)^2 + 1}{2} \int_{\frac{1}{6}}^{\frac{1}{3}} c_1 ds \right] &= \\ \sup_{t \in [\frac{1}{6}, \frac{1}{3}]} \left[\frac{7}{30} + \frac{(R)^2 + 9}{324} \right] &> 0 \end{aligned}$$

and for $i = 2$

$$\begin{aligned} \sup_{t \in [\frac{1}{6}, \frac{1}{3}]} \left[(\beta_2 + \eta_2 - t) \zeta_{2R}^H + \zeta_{2R}^G + \int_{\frac{1}{6}}^{\frac{1}{3}} K_2(t, s) \gamma_{2R}(s) ds \right] &\geq \\ \sup_{t \in [\frac{1}{6}, \frac{1}{3}]} \left[(\beta_2 + \eta_2 - b_2) \frac{1}{10} + \frac{1}{6} + \frac{(\tilde{c}_2 R)^2 + 2}{2} \int_{\frac{1}{6}}^{\frac{1}{3}} c_2 ds \right] &\geq \\ \sup_{t \in [\frac{1}{6}, \frac{1}{3}]} \left[\frac{37}{210} + \frac{9R^2 + 98}{1372} \right] &> 0, \end{aligned}$$

which implies that (7) is satisfied for every $R \in (0, +\infty)$. Thus, we can apply Theorem 2.2 to obtain our desired solution.

3.3 The BVP (2)-(5)

The system (2), together with the set of nonlinear and nonlocal boundary conditions (5) can also be written in the form of system of integral equations (12). Where functions $\psi_{i,0}(t)$, $\psi_{i,1}(t)$ and $K_i(t, s)$ are given by:

$$\begin{aligned} \psi_{1,0}(t) &= \beta_1 + \eta_1 - t \\ \psi_{1,1}(t) &= 1 \\ \psi_{2,0}(t) &= 1 - \frac{t}{\beta_2 + \eta_2}, \\ \psi_{2,1}(t) &= \frac{t}{\beta_2 + \eta_2}, \\ K_1(t, s) &= \beta_1 + (\eta_1 - s) \mathbf{1}_{[0, \eta_1]}(s) - (t - s) \mathbf{1}_{[0, t]}(s), \\ K_2(t, s) &= \frac{t}{\beta_2 + \eta_2} \beta_2 + \frac{t}{\beta_2 + \eta_2} (\eta_2 - s) \mathbf{1}_{[0, \eta_2]}(s) - (t - s) \mathbf{1}_{[0, t]}(s). \end{aligned}$$

The rest is similar to the previous discussion.

Example 3.5. Consider the BVP

$$\begin{cases} -u_1''(t) = \lambda ((u_1(t))^2 + \sin^2(u_2(t)) + 1), \\ -u_2''(t) = \lambda \frac{1}{2} (u_1^2(t) + u_2^2(t) + 1), \\ u_1'(0) + \lambda \left(\frac{1}{10} u_1(1) + \frac{1}{10} u_2(1) + \frac{1}{5} \right) = 0, \\ u_2(0) = \lambda \left(\frac{1}{6} u_1 \left(\frac{1}{3} \right) + \frac{1}{10} u_2(1) + \frac{1}{5} \right), \\ \frac{1}{4} u_1'(1) + u_1 \left(\frac{1}{4} \right) = \lambda \left(\frac{1}{4} \left(\sqrt{u_1 \left(\frac{1}{4} \right)} \right) + \frac{1}{8} (u_2(1)^2) + \frac{1}{5} \right), \\ \frac{1}{3} u_2'(1) + u_2 \left(\frac{1}{4} \right) = \lambda \left(u_1 \left(\frac{1}{3} \right) + u_2 \left(\frac{1}{3} \right) \right). \end{cases} \quad (20)$$

For $R \in (0, +\infty)$, we can take $[a_i, b_i] = \left[\frac{1}{6}, \frac{1}{3} \right]$, $\gamma_{1R}(t) = (\tilde{c}_1 R)^2 + 1$, $\gamma_{2R}(t) = \frac{1}{2}(\tilde{c}_2 R)^2 + 1$, $\tilde{c}_1 = \frac{1}{3}$, $\tilde{c}_2 = \frac{1}{9}$, $\zeta_{1R}^H = 1/5 > 0$, $\zeta_{2R}^H = 1/5$, $\zeta_{1R}^G = 1/5$ and $\zeta_{2R}^G = 0$. Therefore, Condition 3 of Theorem 2.2 is satisfied for $i = 1$,

$$\begin{aligned} \sup_{t \in [\frac{1}{6}, \frac{1}{3}]} \left[(\beta_1 + \eta_1 - t) \zeta_{1R}^H + \zeta_{1R}^G + \int_{\frac{1}{6}}^{\frac{1}{3}} K_1(t, s) \gamma_{1R}(s) ds \right] &\geq \\ \sup_{t \in [\frac{1}{6}, \frac{1}{3}]} \left[(\beta_1 + \eta_1 - b_1) \frac{1}{5} + \frac{1}{5} + \frac{(\tilde{c}_1 R)^2 + 1}{2} \int_{\frac{1}{6}}^{\frac{1}{3}} c_1 ds \right] &= \\ \sup_{t \in [\frac{1}{6}, \frac{1}{3}]} \left[\frac{7}{30} + \frac{(R)^2 + 9}{324} \right] &> 0 \end{aligned}$$

and $i = 2$

$$\begin{aligned} \sup_{t \in [\frac{1}{6}, \frac{1}{3}]} \left[\left(1 - \frac{t}{\beta_2 + \eta_2} \right) \frac{1}{5} + \frac{1}{2} ((\tilde{c}_2 R)^2 + 1) \int_{\frac{1}{6}}^{\frac{1}{3}} K_2(t, s) ds \right] &\geq \\ \sup_{t \in [\frac{1}{6}, \frac{1}{3}]} \left[\left(\frac{4}{35} \right) + \frac{1}{162} (R^2 + 1) \int_{\frac{1}{6}}^{\frac{1}{3}} c_2 s ds \right] &= \\ \sup_{t \in [\frac{1}{6}, \frac{1}{3}]} \left[\frac{4}{35} + \frac{3}{17496} (R^2 + 1) \right] &> 0, \end{aligned}$$

A similar result as the ones above also holds in this example.

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